ESTIMATION OF GLACIER ICE THICKNESS CHANGE AND MASS BALANCE USING GEODETIC METHOD

Master of Technology Project Report

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Abstract

The Himalayas which has the highest concentration of glaciers outside the poles forms a huge reservoir of fresh water. These glaciers are not only important because of the contribution they make to our water resources, but also because they are a great help in understanding the effects of climate change. The Indian Himalayas have more than 9500 glaciers which contribute to the country's water resources. Any change in glacier size and volume will have serious impact on water management. Therefore, there is a great interest and high demand on definite quantification of changes in Himalayan glaciers However, monitoring even a few of these glaciers through field techniques is a tedious process due to the tough and unapproachable terrain of the Himalayas. In such situation, remote sensing techniques becomes a boon in studying glaciers which are unapproachable. In this thesis one such widely used remote sensing based technique called Digital Elevation Model (DEM) Differencing or Geodetic approach available for estimating glacier surface mass balance (SMB) is discussed. However, the accuracy of the estimated surface mass balance using geodetic approach is highly dependent on the accuracy of the DEMs used, which utlimately decides whether this approach can be used for seasonal, annual or decadal studies. To get a better understanding of this approach, this thesis gives a overview of errors present in DEMs and how to correct it before estimating suface mass balance by demonstrating it over 65 glaciers located in the Chandra basin, Lahaul-Spiti district of Himachal Pradesh, India. The obtained results in the study indicate that the Chandra basin has lost 5.01 ± 2.48 Gt of ice mass during the course of 14 years (1999-2013), which gives a clear signal on the effect of climate change occurred over the study area and how climate change could cause imbalance in the health of glacier(s). Considering the advantages of this approach and recent developments in high resolution DEM data acquisition as well as the need for such accurate mass balance estimates across Himalayas, it is therefore suggested that this modelling approach should be extended to other basins to get the current status of the health of the glaciers across Himalayas.

Contents

Abstract i

List of Figures iv

List of Tables v

Abbreviations vi

1 Introduction 1

1.1 Motivation of the study 1

1.2 Outline of the report 2

1.3 Aim and objectives 2

2 Literature review 3

2.1 Review of state-of-the-art glacier mass balance studies using geodetic method 5

3 Study area and Datasets 8

3.1 Study area description 8

3.2 Datasets used 9

3.2.1 Glacier outlines 9

3.2.2 DEMs 9

3.2.3 Landsat data 11

4 Methods 12

4.1 Pre-processing 13

4.2 Penetration correction 13

4.3 DEM Co-registration 14

4.4 Resolution bias correction 15

4.5 Mass Balance calculation 16

4.6 Seasonality correction 17

4.7 Uncertainty assessment 17

5 Results 19

5.1 Results for Co-registration of DEMs 19

5.2 Results for resolution bias correction 20

5.3 Volume change and mass balance 1999 - 2011 22

5.4 Volume change and mass balance 1999 - 2013 23

5.5 Comparison of results..... 24

6 Conclusions..... 26

References..... 27

Acknowledgement 32

List of Figures

Figure 3-1Map of the study area showing the locations of the selected 65 glaciers in Chandra river basin, Lahaul-Spiti valley, Himachal Pradesh, India. 8

Figure 4-1 Flow chart..... 12

Figure 4-2: 2-D scheme of elevation differences induced by a DEM shift 15

Figure 4-3 Relation between altitude and elevation difference 16

Figure 5-1 The scatter of elevation differences between 2 DEMs showing the relationship between the vertical deviations normalized by the slope tangent (y-axis) and terrain aspect (x-axis). (a) TanDEM-X – SRTM. (b) ASTER - SRTM..... 20

Figure 5-2 Relation between altitude and elevation difference before and after resolution bias correction 21

Figure 5-3 elevation change map of all the glaciers in Chandra basin between February 2000 and October 2013..... 21

Figure 5-4 elevation change map of all the glaciers in Chandra basin between February 2000 and October 2011..... 22

Figure 5-5 Specific mass balance rate of selected 65 glaciers (1999-2011)..... 23

Figure 5-6 Specific mass balance rate of selected 65 glaciers (1999-2013)..... 24

List of Tables

Table 3-1Details of the TanDEM-X data used for DEM generation.....11

Table 3-2 Details of Landsat images used 11

Table 4-1 Penetration of SRTM C-band radar waves as estimated from extrapolating the ICESat elevation difference trends to the SRTM acquisition time in February 2000(Kaab et al., 2012). 14

Table 5-1 Comparison of Specific SMB rates from present study with other studies in Chandra basin 25

Abbreviations

AAR	Accumulation Area Ratio
ALOS	Advanced Land Observing Satellite
ASTER	Advanced Spaceborne Thermal Emission and Reflection Radiometer
DEM	Digital Elevation Model
ELA	Equilibrium Line Altitude
ETM+	Enhanced Thematic Mapper +
GMB	Glacier Mass Balance
HKH	Hindu Kush Himalayan
IPCC	Inter-Governmental Panel on Climate Change
PRISM	Panchromatic Remote-sensing Instrument for Stereo Mapping
PTAA	Precipitation Temperature Area Altitude
SMB	Surface Mass Balance
SPOT	Satellite Pour l'Observation de la Terre
SRTM	Shuttle Radar Topography Mission
TanDEM-X	TerraSAR-X Add-On for Digital Elevation Measurement
UAV	Unmanned Aerial Vehicle

1 Introduction

1.1 Motivation of the study

The Himalayas have the highest concentration of glaciers outside the poles, which makes them a huge source of fresh water for the developing world to which they feed. One third of the humanity on this planet is directly or indirectly dependent on the nine major river systems to which headwater is provided by the Himalayan glaciers. The runoff coming from these glaciers not only affects the drinking water supplies, but also the hydro power projects, biodiversity, industries, agriculture and day to day life of dependent individuals among others. These glaciers are not only important because of the contribution they make to our water bodies, but also because they are a great help in studying climate change. Some scientists also refer to these glaciers as thermometers which can be used to study the long-term impacts of climate change.

Crucial to the survival of a glacier is its mass balance (GMB), the difference between accumulation and ablation (melting and sublimation). There is an overwhelming evidence that the Himalayan glaciers are not in the best of health and most of them are losing mass. While most of the glaciers are losing mass at a rate similar to glaciers at other places, Karakoram Himalayas emerge out as an exception with some studies suggesting mass gain (Bolch et al., 2012). Climate change may cause variations in both temperature and snowfall, causing changes in mass balance. Changes in mass balance control a glacier's long term behaviour. It is the most sensitive climate indicator on a glacier. A glacier with a sustained negative balance is out of equilibrium and will retreat. A glacier with a sustained positive balance is out of equilibrium and will advance.

Glaciers that are monitored routinely are often chosen more for their ease of access and manageable size than for how well they represent a given region or how large a contribution they might make to changing sea level. As a result, global mass balance data are biased toward small glaciers (20 km^2) rather than those that contain the most ice (100 km^2). Also, large cumulative errors can result from using only a few point measurements to estimate glacier-wide mass balances on an individual glacier (Arendt et al., 2002).

Mountain glaciers are also widely recognized to be sensitive climatic indicators (Houghton et al., 2001). Measuring glacier evolution gives insights into the regional climate change in high and remote places where meteorological measurements are difficult and rare. This is of particular interest in the western Himalaya because the region is influenced by two major climatic systems (the mid-latitude Westerlies and the south Asian summer monsoon) which are

still poorly known due to logistical difficulties in maintaining observational networks at high elevations (Wagnon et al., 2007).

1.2 Outline of the report

Chapter 1 discusses the importance of monitoring glacier mass balance in the current rapidly changing climate. Based on these, the aim and research objectives for the present study have been enlisted.

Chapter 2 consists of the basis of glaciers mass balance. The approaches used to estimate glacier mass balance have also been discussed in this chapter. Then some studies in estimation of glacier mass balance using geodetic method are also discussed.

Chapter 3 discusses the study area selected form the study, available datasets.

Chapter 4 discusses the proposed methodology for geodetic mass balance of Glaciers.

Chapter 5 discusses the results of the study based on the proposed methodology.

Chapter 6 summaries the report with the future works to be done.

1.3 Aim and objectives

The main aim of the present study is to estimate the glacier ice thickness change and mass balance of glaciers using geodetic method. Geodetic method is chosen in this study as it uses minimum amount of data and provides an accurate estimate over large area which is also proven in other studies. This is applied on Chandra Basin of Western Himalayas.

The objectives of the study are to:

- Apply required corrections to DEMs.
- Estimate the ice thickness change of glaciers in Chandra basin.
- Evaluate glacier mass balance of glaciers in Chandra basin.
- Compare the results of the study with results of other related studies.

2 Literature review

Different methods are available that determine the ice-thickness change distribution and total volume loss/gain of the mountain glaciers. These methods are based on, for example, in-situ methods, simple physical models, empirical relationships or topographic maps.

The glaciological method is the only which is based on in situ measurements. At a number of individual points, the change in surface level is measured between two dates. The difference in level (gain or loss), multiplied by near-surface density, yields an estimate of the mass balance at that point. Changes in level are measured in a variety of ways, including stakes drilled into the glacier time (Dobhal et al., 2004; Kumar et al., 2009) and snow depths relative to a known stratigraphic surface (e.g. previous summer surface). Density values for ice are assumed constant at 900 kgm^{-3}. Snow density is measured in snow pits, which are dug down to a reference surface. Density can also be measured from cores taken with a drill. Monitoring the glacier mass balance by glaciological technique of observing stakes placed over a glacier's surface is difficult and is limited in scope. As most of the glaciers are located in the non-accessible and remote areas where terrain is very rigid and complex, and weather is so extreme and harsh, making the task of collecting data for mass balance estimation difficult.

Next comes the hydrological method which calculates glacier mass balance as a storage term in the water balance. Subtraction of runoff and evapotranspiration from precipitation delivers the glacier mass budget for the whole catchment area (Hagg et. al., 2004). From a hydrological point of view, a glacier acts as a reservoir with seasonal gains and losses. Thus, the glacier mass balance is one term in the hydrological balance of a glaciered catchment basin and can be calculated as a residual of all other terms:

$$B = P - Q - E \pm \Delta S \tag{2.1}$$

Where, P= precipitation; Q= runoff; E= evaporation; ΔS= variation of storage elements of the catchment area other than glaciers such as groundwater or interception. This method requires good instrumentation to measure each of the variables. This is a challenging effort for unattended operation in high alpine basins. Maintaining a good gauging station for water discharge can be an expensive and time- intensive operation itself, also gauging stations near glaciers are rare. Also, extrapolation of precipitation from a single gauge to the surrounding mountainous terrain is often inaccurate. Finally, the natural processes of storage and release of water within a glacier can confound this method. Typically, the inaccuracies of the measured variables equal the order of magnitude of B. The hydrological method is used to compute the

mass balance of Siachen Glacier in the Nubra Valley, Karakoram Himalaya, India by Bhutiyani 1999. Due to the need of extensive instrumentation required for carrying out the hydrological method, this method is not preferred.

Another way of calculating mass balance is modelling from climate records. The mass balance of a glacier results from climate. If non- climate mechanical processes can be excluded (e.g. surging, kinematic waves, avalanches) the mass balance can be derived from climate records. Depending on the accuracy and availability of climate data a variety of models can be applied. Most models focus on the ablation season using simple degree-day approaches or more sophisticated energy balance methods. Because of the complexity of precipitation distribution and accumulation, most models rely on a simple extrapolation from precipitation data. Tangborn, 1997 developed a model called PTAA (Precipitation Temperature Area Altitude) which used the precipitation and temperature data from a nearby low altitude weather station and the area- altitude distribution of a glacier to compute various glacier mass balance parameters at daily scale. The model has been applied to Columbia glacier, Alaska (Tangborn, 1997) and South Cascade glacier (Tangborn, 1999). To date, all of models have to be calibrated for the glacier in question. Thus, mass balance data are needed, at least in the beginning (Kesar et al., 2003).

Next are the methods which use remotely sensed data as input and calculate the mass balance. The Accumulation Area Ratio/Equilibrium line altitude (AAR/ELA) method for mass balance estimation was proposed by Kulkarni, 1992. It is an indirect method of calculating the mass balance of a glacier. The method involves establishing an empirical relation between AAR and mass balance. He has shown that if long term field data of glacier is available, a linear regression relation can be established between AAR and mass balance of the form

$$B_s = a * AAR + b \qquad (2.2)$$

Where B_s is the specific mass balance in water equivalent (m) and AAR is the accumulation area ration, *a* and *b* are constants. The linear equation can be used to predict the mass balance of the glacier by calculating AAR from remote sensing data analysis. According to Kulkarni, 1992, this equation can also be used to estimate the mass balance of other glaciers located in the same climatic zone if the AAR from remote sensing data is available. This method was applied to Gara and Gor Garang glaciers in Himachal Pradesh using field data from various sources. The regression analysis established has found a generalized value of AAR of 0.44 for western Himalaya (Kulkarni, 1992).

Last of all there is geodetic method. In which the volume change can be estimated by subtracting the surface elevation of a glacier and the glacier extent at two different times. Knowing the surface density at different parts of the glacier the volume change can be converted into a mass change. This method can be applied using topographic maps, digital elevation models obtained by aircraft and satellite imagery, and by airborne laser scanning (Kesar et al., 2003).

Since last two decades, many of the studies has been done which include snow cover mapping, mass balance measurements, monitoring of snout, meteorological measurement studies, glacier hydrology including discharge measurement, runoff measurement, sediment load and water quality, using multispectral remote sensing as well as ground penetrating radar, but all these studies are restricted over small geographical areas and for short periods. Geodetic method is one of the newest techniques which is being used to calculate mass balance of glaciers. This method has several advantages over conventional methods like glaciological and hydrological methods. Using geodetic method, we can overcome geographical barriers and access remote glaciers. This reduces human labour and life risk. The method can also be applied to large number of glaciers and doesn't need to be recalibrated for individual glaciers. Hence, this method can be used for carrying out studies over a large geographical area.

2.1 Review of state-of-the-art glacier mass balance studies using geodetic method

This section discusses about the research carried on the estimation of glacier mass balance using geodetic method.

In a comparative study of different methods of mass balance measurement done for the Tuyuksu glacier region in the northern Tien Shan, Central Asia, good correlation was found between geodetic method and field glaciological measurements (Hagg et al., 2004, Rabatel et al., 2005).

Berthier et al., 2004 have derived thickness change of Mer de Glace of French Alps using different year's satellite elevation data and found rapid thinning of the glacier below 2500m. In another study of elevation change and geodetic mass balance, Berthier et al., 2010 have quantified the contribution of Alaskan glacier in sea level rise using satellite imageries. In the study they have applied the sequential digital elevation model (DEM) analysis to estimate the mass loss of Alaskan glaciers over the period 1962-2006. They calculate ice elevation changes for nearly three quarters of the ice-covered areas of Alaska by subtracting an old DEM derived

from map elevation contour lines from a recent DEM derived from SPOT5 and ASTER images. They found that between 1962 and 2006, Alaskan glacier lost 41.9±8.6 km^3 $year^{-1}$ of water and contribute 0.12±0.02 mm $year^{-1}$ to seas level rise.

For the Himalayan glaciers also, the elevation and volume changes have been estimated using satellite elevation data and geodetic method. Berthier et al., 2007 have used SRTM and SPOT 5 DEM to determine elevation change in Himachal glaciers during 1999 and 2004 by comparing the two DEMs. They have reported that between 1999 and 2004 an overall specific mass balance of - 0.7 to -0.85 m we y^{-1}depending on the density used for the lost (or gained) materials in accumulation zone.

Studies have also been done on Khumbu region, Himalaya to estimate the volume and elevation change (Pieczonka et al., 2011, Nuimura et al., 2012). Pieczonka et al., 2011 have used DEM derived from CORONA KH4 1962 and ASTER DEM 2002 for volumetric change study of Khumbu region. Their results show an average thickness loss of about 17±9 m corresponding to 0.4±0.2 m y^{-1} between 1962 and 2002.

Bolch et al., 2011 have derived multi decadal mass loss of glacier in Everest area derived from stereo imageries using geodetic method. They have used CORONA spy images of 1962 and 1970, aerial images and high-resolution images of Cartosat-1 to reconstruct longest time series of mass changes in the Himalaya. They reported that despite of heavy debris cover, the glaciers have been losing mass significantly since 1970. The rate of mass loss has increased over the years. They report a specific mass loss of 0.32±0.008 m we. y^{-1} for 1970-2007.

Nuimura et al., 2012 have derived elevation change of the Khumbu region between 1992 and 2008 by using multi-temporal DEMs. Their study confirms the elevation change of glaciers in the eastern Khumbu region. They report a regional average mass balance of -0.40±0.25 m we. y^{-1} for the period 1992- 2008.

Gardelle et al., 2012 have utilized SRTM and SPOT 5 data for studying central Karakoram glaciers using geodetic method between 1999 and 2008. Results shown small positive mass balance of Karakoram glaciers at 0.11±0.22 m we. y^{-1}

Vincent et al., 2013 has extend the Chhota Shigri MB back to 1988 using in situ geodetic measurements performed in 1988 and 2010. Using satellite DEM differencing and field measurements, they measure a negative mass balance between 1999 and 2010 (–0.44±0.16 m

we y^{-1}) and a slightly positive or near-zero Mass Balance between 1988 and 1999 (+0.09±0.24 m we y^{-1}).

King et al., 2017 In their study, they quantified the mass loss rates over the period 2000–2015 for 32 glaciers across the Everest region using Geodetic method and assess how future ice loss is likely to differ depending on glacier hypsometry. The mean mass balance of all 32 glaciers in their sample was -0.52±0.22 m we.

Ruiz et al., 2017 study presents a geodetic mass balance estimate of Monte Tronador glaciers by comparing a Pléiades DEM acquired in 2012 with the Shuttle Radar Topography Mission (SRTM) X-band DEM acquired in 2000. They found a slightly negative Monte-Tronador-wide mass budget of -0.17 m we over the 2000–2012 period. With a few exceptions, debris-covered valley glaciers that descend below a bedrock cliff are losing mass at higher rates, while mountain glaciers with termini located above this cliff are closer to mass equilibrium.

Vijay and Braun 2016 present a study on elevation changes of glaciers in the Lahaul-Spiti region derived from TanDEM-X and SRTM C-/X-band DEMs during 2000–2012 and 2012–2013. They observed three elevation change patterns during 2000–2012 among glaciers with different extent of supraglacial debris.. They empirically determined the SRTM C- and X-band penetration differences for debris covered ice, clean ice/firn/snow. They show that this penetration bias, if uncorrected, underestimates the region-wide elevation change and geodetic mass balance by 20%.

Accuracy of geodetic method has several issues. Its depends highly on the accuracy of DEMs used, discrepancies in scale of resolution of DEMs, the assumption of density of lost/gained material (snow/firn/ice) and inherent biases in the remote sensing derived DEMs (Racoviteanu et al., 2007). This method is suitable for estimation of long term variation (5-10 years) rather than yearly variation over glacier area because of the low vertical accuracy of generated DEMs

3 Study area and Datasets

3.1 Study area description

The present study is carried out on Chandra Basin which is a sub basin of the Chenab basin, which is one of the most important river basins of India because it caters to the needs of millions of people in north India and the water of this river is also shared with Pakistan. It is located in Lahul and Spiti district of Himachal Pradesh in India (Figure 3-1). The spatial extent of the area is 69:8^0 E to 77:5^0 E and 32^0 N to 33^{0}N. This basin lies on the northern ridge of Pir Panjal range of the Himalayas with an elevation range lying between 2400 m above mean sea level to 6400 m above mean sea level.

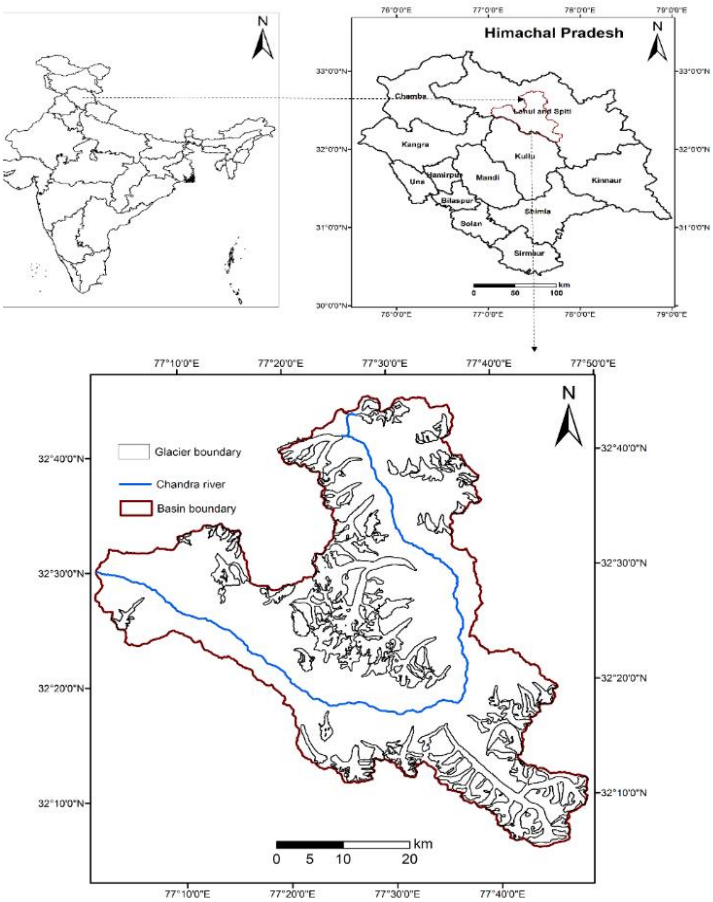

Figure 3-1Map of the study area showing the locations of the selected 65 glaciers in Chandra river basin, Lahaul-Spiti valley, Himachal Pradesh, India.

Glaciers in this region are fed by South Asian monsoons in the summers and western disturbance (westerlies) in the winters which makes them even more important for climate change studies. High wet precipitation is recorded in summer (July - September); however,

winter (November-February) experiences a significant amount of solid precipitation due to the influence of the westerlies (Patel et al., 2017).

The Chandra basin supports 201 glaciers and glaciates covering a net area of 703.64 km^2, which constitutes 29.54% of the total basin area. In this study, some 65 glaciers of Chandra basin have been selected surface mass balance estimations which covers an area of 535.4 km^2. Glaciers less than 0.5 km^2 are not considered in this research because generally smaller glaciers consist of large uncertainties in their area due to difficulty in glacier outline delineation

3.2 Datasets used

3.2.1 Glacier outlines

Outline of the study glaciers was obtained from the Randolph Glacier Inventory 6.0 (https://www.glims.org/RGI/) and wherever required, minor corrections like adjustment in edges of the glacier boundary have been done using Google Earth images. The Randolph Glacier Inventory (RGI) is a collection of digital outlines of the world's glaciers, excluding the Greenland and Antarctic ice sheets. The RGI was developed in a short (1–2 year) period with limited resources in order to meet the needs of the Fifth Assessment Report (AR5) of the Intergovernmental Panel on Climate Change (IPCC) for estimates of recent and future glacier mass balance. Priority was given to complete coverage rather than to extensive documentary detail (Pfeffer et al., 2014).

Most of the glacier outlines in the RGI with known dates were derived from satellite imagery acquired in 1999 or later. Various Landsat platforms, principally Landsat 5 TM and Landsat 7 ETM+, were the primary sources, and imagery from ASTER, IKONOS and SPOT 5 high-resolution stereo (HRS) sensors was also used. Each object in the RGI conforms to the data-model conventions of ESRI ArcGIS shapefiles.

3.2.2 DEMs

In present study we used all the DEMs available freely for in our study area. The geodetic mass balance estimates are calculated using SRTM, ASTER and TanDEM-X DEMs. The SRTM V3 was obtained from USGS earth explorer (http://earthexplorer.usgs.gov/) and ASTER DEM (AST14DMO) from NASA Land Processes Distributed Active Archive Center(https://lpdaac.usgs.gov/dataset_discovery/aster/aster_products_table/ast14dmo_v003). whereas TanDEM-X DEM is generated using TanDEM-X co-registered Single Look Slant Range Complex (CoSSC) product which was obtained under TanDEM-X Science proposal XTI_GLAC7043. Other DEMs like CartoSat, high resolution Advanced Land Observing

Satellite (ALOS) Panchromatic Remote-sensing Instrument for Stereo Mapping (PRISM) and High Mountain Asia (HMA) DEMs are also freely available for our study area but are not used for many reasons. Most of our study glaciers lie in the voids of the HMA DEM. The Acquisition date of the ALOS PRISM DEM is not known. Coming to CartoSat DEM since it an optical based DEM Cloud cover there were many outliers in the DEM. Also, studies proved that CartoSat has poor vertical accuracy in mountainous terrain (Singh et al., 2016).

3.2.2.1 SRTM DEM

The Shuttle Radar Topography Mission (SRTM) is a joint program between NASA and National Geo-spatial intelligence Agency (NGA) to map the globe in 3 dimensions. SRTM utilized dual Spaceborne Imaging Radar (SIR-C) and dual X-band Synthetic Aperture Radar (X-SAR) configured as a baseline interferometer, acquiring two images at the same time. These images when combined, can produce a 3-dimensional image. Flown on 11– 22 February 2000, SRTM had the first space borne single-pass SAR interferometer. It yielded single-pass interferometry data, C and X-Bands, of land between 60N and 54S for processing into global DEMs. The InSAR antenna configuration (precise measurement of relative antenna location with laser rangers, star trackers and in-orbit Global Positioning System - GPS) resulted in elimination of mapping errors characteristic of repeat-pass satellite InSAR techniques. The nominal vertical accuracy is 6 m relative, 16 m absolute, and the nominal horizontal accuracy is 15 m relative, 20 m absolute, at the 90% confidence level (Muskett et al. 2003).

3.2.2.2 ASTER DEM

The Advanced Spaceborne Thermal Emission and Reflection Radiometer (ASTER) Global Digital Elevation Model Version 2 (GDEM V2) was jointly announced on October 17, 2011 by The Ministry of Economy, Trade, and Industry (METI) of Japan and the United States National Aeronautics and Space Administration (NASA). The ASTER GDEM V2 maintains the GeoTIFF format and has a resolution of 30m and covers land from 83ºN to 83ºS.

3.2.2.3 TanDEM-X DEM

TanDEM-X (TerraSAR-X add on for Digital Elevation Model) is an innovative space borne radar interferometer that is based on two high-resolution radar satellites in close formation flight: TerraSAR-X (launched on June 15, 2007) and TanDEM-X (launched on June 21, 2010). The primary objective of the TanDEM-X mission is the generation of a consistent global digital elevation model with high accuracy.

Details about the TanDEM-X satellite data used in this study for DEM generation are given in Table 3-1.

Table 3-1Details of the TanDEM-X data used for DEM generation

S. No.	Study Area	Date of acquisition	Polarization	Normal Baseline (m)	Pass	Incidence angle (deg.)
1	Scene 1	27th Oct 2013	HH	182.66	Descending	34.39°
2	Scene 2	27th Oct 2013	HH	182.72	Descending	34.56°

TanDEM-X DEM is generated using the methodology given in *Pandit et al., 2014*

3.2.3 Landsat data

For the present study, Landsat data is used to map different facies of the glacier. Cloud-free 30-m resolution Landsat TM or ETM scenes of the entire study region from around the year 2000 with minimal snow cover are obtained from the USGS in topographically corrected version (L1T). The details of the images used are given in Table 3-2

Table 3-2 Details of Landsat images used

Landsat series	**Image ID**	**Date of acquisition**	**Path**	**Row**
Landsat 7	LE07_L1TP_147037_20001015_20170209_01_T1	15/10/2000	147	38
Landsat 7	LE07_L1TP_147038_20001015_20170209_01_T1	15/10/2000	147	37

4 Methods

Based on the availability of data the following methodology has been selected (Figure 4-1). Flow chart shows the detail framework of the present study, which gives broad steps involved in completing the study. The section also include various corrections required for the DEMs, reasons for the correction and the method adopted for correction. Various corrections in the section are explained in the same hierarchy shown in the flow chart (Figure 4-1)

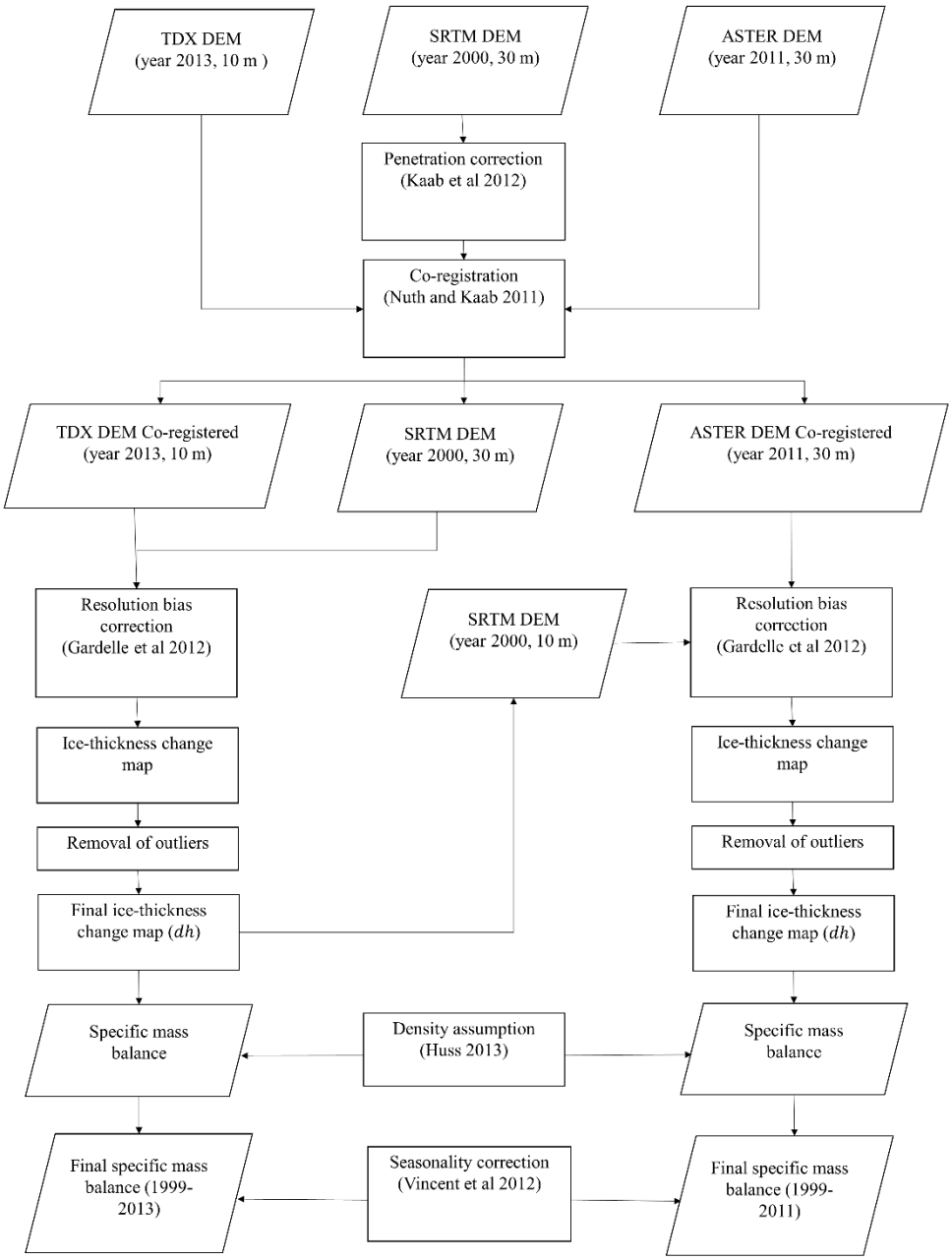

Figure 4-1 Flow chart of the methodology adopted for calculating mass balance

4.1 Pre-processing

Since we are using DEMs from different sources we have to make sure that all the DEMs have same spatial projection and vertical datum. In our case we are projecting all the DEMs in UTM WGS84, Zone 43 and geoidal heights are used to calculate the elevation difference. The spatial projections are changed using ArcMap10.2.2. ASTER DEM when downloaded gives ellipsoidal heights. Therefore, these ellipsoidal heights are converted into geoidal heights using ENVI 5.1 SARScape. Once all the DEMs are in same projections we can proceed into further steps.

4.2 Penetration correction

The C-Band radar penetration of SRTM can reach up to 10 m in Snow and Ice. The DEM thus derived maps a surface below real surface, especially in accumulation areas, leading to biased estimates of glacier elevation changes. Therefore, a correction in the elevations of SRTM DEM are needed for correct estimation of elevation changes.

From literature it is seen that two methods are generally used to make this correction one of which involves the SRTM-X band data (Gardelle et al., 2012b) and the other uses the ICESat data (Kääb et al., 2012). In present study we adopted the method used in Kääb et al., 2012 as the SRTM-X band data for our study area is not available.

Kääb et al., 2012 assessed the magnitude of C-band penetration over various test sites in the Himalayas and over different ice facies (clean ice, snow and firn) by extrapolating ICESat vs. SRTM glacier elevation differences back to the SRTM acquisition date, showing penetration estimates of several metres. As the ICESat data is available only from 2003 the ICESat data is extended back to 2000 (SRTM Acquisition time) by linear extrapolation. This linear extrapolation is however based on the assumption that temporal changes in *dh* trends between 2000-2003 and 2003-2009 are negligible. Giving penetration values over different facies at different study sites like Hindu-Kush (HK), Karakoram (KK), Jammu Kashmir (JK), Himachal Pradesh, Nepal, Uttarakhand and Bhutan (see Table S2 of Kääb et al., 2012). The penetration values for our study area are given in Table 4-1

Table 4-1 Penetration of SRTM C-band radar waves as estimated from extrapolating the ICESat elevation difference trends to the SRTM acquisition time in February 2000(Kaab et al., 2012).

SRTM Penetration (m)	Hindu-Kush (HK)	Karakoram (KK)	Jammu Kashmir (JK)	Himachal Pradesh (HP)	East Nepal and Bhutan (NB)	Area Weighted mean (HKKH)
Off-Glacier	0.1±0.4	-0.2±0.2	0.6±0.2	0.1±0.2	-0.2±0.2	0.0±0.2
Glacier	2.4±0.4	2.4±0.3	1.4±0.6	1.5±0.4	2.5±0.5	2.1±0.4
Firn/Snow	5.1±0.7	5.5±0.3	2.3±0.9	2.3±0.6	4.8±0.7	4.3±0.5
Clean Ice	1.7±0.6	1.1±0.5	1.7±0.7	1.7±0.6	0.1±1.2	1.2±0.6
Debri covered Ice	2.0±0.8	-2.9±0.9	2.0±1.1	0.4±0.6	-0.8±0.1	-0.6±0.9

To perform the correction adopted various facies on the glacier should be mapped. For mapping the glacier facies, we used the method given in Kääb et al., 2012. Firstly Landsat 7 band3/band5 ratio larger than 2.2 is used to separate snow, firn and clean ice footprints from the Debris and non-glacier foot prints. Then, Firn and snow are discriminated from clean ice by a threshold of ~200 on the ratio (band4×band2)/band5 after stretching to values from 0 to 255. Finally, Footprints over debris-covered glacier sections are marked using glacier inventory outlines.

4.3 DEM Co-registration

DEM co-registration as the name suggests means to align different DEM datasets. As the misalignment of DEM could result in wrong elevation change estimates it is crucial to correct. Moreover, errors and biases may persist from sensor instabilities, limitations of the techniques, and post-processing (Nuth and Kääb, 2011). Figure 4-2 shows two DEMs which are not perfectly aligned. Therefore, the resulting elevations changes (dh) are larger on steeper slopes, positive on eastern slopes and negative on western slopes. Understanding and handling these misalignments are necessary when planning to research on the elevation change estimates.

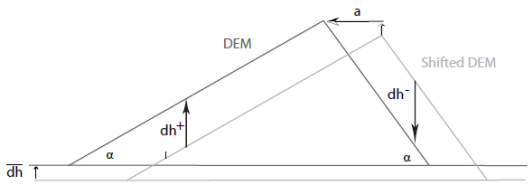

Figure 4-2: 2-D scheme of elevation differences induced by a DEM shift

This study follows the method explained by Nuth and Kääb, 2011. This method is adopted as the method turned out to be sufficient and efficient from the day it was published. Many other studies like Gardelle et al., 2012b, Melkonian et al., 2013, Paul et al., 2012, Gardelle et al., 2013, Pieczonka et al., 2013, Karimi et al., 2012, Nuth et al., 2013, in this study the method explained by Nuth and Kääb, 2011 has been adopted for co-registration correction.

In universal co-registration tool developed by Nuth and Kääb, 2011, which establishes an analytical relationship (equation 4.1,4.2) between the dh, terrain aspect (Ψ) and slope (α). The analytical equation is solved using least square minimization (Nuth and Kääb, 2011) until certain conditions are met (i.e., standard deviation threshold, improvement of the standard deviation is less than 2%) to calculate the shift vectors (Nuth and Kääb, 2011).

$$\frac{dh}{\tan(\propto)} = a\,.\cos(b - \Psi) + c \tag{4.1}$$

Where,

$$c = \frac{\overline{dh}}{\tan(\overline{\propto})} \tag{4.2}$$

4.4 Resolution bias correction

A bias in elevation changes is observed when the DEMs are subtracted and plotted with the altitudes (Berthier et al., 2006 and Berthier et al., 2007). This bias in elevation difference with increase in altitude can be clearly observed in present study (Figure 4-3). This bias has been corrected in many studies (Surazakov and Aizen, 2006; Schiefer et al., 2007; Nuth et al., 2011), Although the reason of this bias was not clearly known some studies have suggested reasons like varying resolution (Paul, 2008), more penetration of SRTM-C band at higher elevations (Berthier et al., 2006). Either way, an elevation dependent bias is extremely significant for estimating glacier volume changes because a glacier and its mass balance varies largely with elevation. However, we cannot correct this bias directly because this bias is not similar on and off the glaciers (Gardelle et al., 2012).

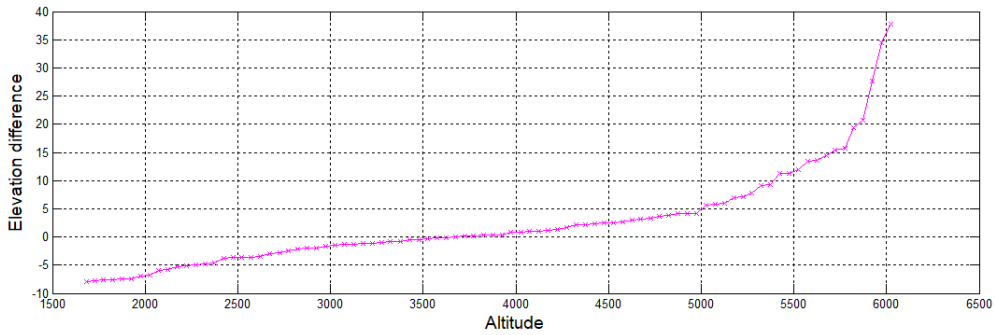

Figure 4-3 Relation between altitude and elevation difference

One of the known reason for this bias is due to the varying resolutions of the DEMs used. To explain this let's see how slopes are represented by low and high resolution DEMs. A slope constant over numerous continuous pixels will be equally represented by low and high resolution DEMs, on the opposite, a slope that is constant only over few high-resolution pixels will be better represented by high resolution DEM than by the coarser DEM (Gardelle et al., 2012). This establishes a relationship between change in slope (curvature) and elevation difference and it is found that elevation differences have strong bias with the maximum curvature. In present study the maximum curvature (for high resolution DEM) is calculated with a kernel size of 5X5 the computation is done in ENVI software. A first-degree polynomial is used to correct the bias in elevation. As the curvature varies on the glacier the correction applies local changes with effect each glacier in a different way depending upon their curvature distributions.

4.5 Mass Balance calculation

To calculate the Mass balance from the bias corrected ice thickness change map, the elevation changes on a glacier that exceed ±100 m are discarded initially, which are surely considered to be outliers (Berthier et al.,2004). Next, in order to eliminate the further outliers present in the thickness change, we have discarded pixels for which the height difference differed by more than 3 standard deviations (±100 m) from the mean. Then, using a 3x3 averaging filter the excluded pixels were filled. The filled ice thickness map was then converted into specific SMB using Equation 4.3 in which the density of ice is assumed to be 850 ± 60 kg/m^3 as adopted by Huss 2013.

$$M = \frac{\int_A dh}{A} \times \frac{\rho_{ice}}{\rho_{water}} \tag{4.3}$$

Where, M = specific SMB (m w.e), dh = Elevation difference (m), ρ_{ice}=Density of ice (kg/m^3), ρ_{water} = Density of water (1000 kg/m^3), A = Glacier surface area (m^2)

4.6 Seasonality correction

When the survey dates are not separated by an integer number of years seasonality corrections are necessary. They restore the elevations on the glacier to the end of the balanced year or any other survey date. These corrections are assumed to be in significant when measured thickness changes are large by comparison with the amplitude of the seasonal cycle of thickness (Cogley, 2009). In present study SRTM was obtained in Mid-February which is accumulation period and TanDEM-X and ASTER are obtained in October which is the end of ablation period. Therefore, if seasonality correction is not applied it will over-estimate the elevation changes.

The correction is applied using the seasonal mass balance of glaciers in the study area. In our case to make the survey an integer the SRTM DEM acquired February in 2000 is shifted to October 1999. Therefore, the accumulation in this 4-month period is subtracted from the SRTM DEM

4.7 Uncertainty assessment

Errors in mass balance calculation comes from different sources. In this study we consider the uncertainty in elevation change ($E_{\Delta h}$), uncertainty in the density assumption (E_{den}) and uncertainty in area (E_{area}).

Major uncertainty in elevation change comes from the errors in individual elevation pixels of the DEM. It is assumed that the error in elevation change is equal to the standard deviation of the mean elevation changes on the glacier (σ_{dh}). As spatial correlation usually exists in elevation data the error in elevation change E_{dh} is expressed as:

$$E_{dh} = \frac{\sigma_{dh}}{\sqrt{N_{eff}}} \tag{4.4}$$

N_{eff} represents the effective number of measurements. N_{eff} will be lower than N_{tot} ,the total number of elevation measurements on the glacier. N_{eff} can be calculated by:

$$N_{eff} = \frac{N_{tot}.PS}{2d} \tag{4.4}$$

Where PS is the pixel size, d is the distance of spatial auto correlation calculated using Moran's I auto-correlation index of elevation differences on stable terrain. In this study we calculated Moran's I using ENVI software. To calculate d, we start with the map of elevation change. Then we create a mask to exclude unreliable data (outside the reasonable data ±100) and data

on the glacier surface. Gardelle et al., 2013 calculated the value of d as lag distance of pixels for which the Moran's I reach $\frac{1}{e}$ i.e., 0.368. The same approach is used in this study and the value of d is calculated to be 767 m for the study between 1999 and 2011 and 725 m for study between 1999 and 2013. The value of d is calculated over 9 different Sites and the average value of the 9 sites is considered in the study.

The uncertainty in elevation change is also caused due to the penetration correction and seasonality correction. The error due to penetration correction is assumed to be systematic and equal to ±1.5m (Gardelle et al., 2013). The uncertainty due to seasonality correction was assumed 100% that is 0.15 per winter month. In our case it is 4-months summing up to a total error of ±0.6m. All these errors in elevation change are summed up for each glacier using error propagation. This summed up error is denoted as $E_{\Delta h}$

$$E_{\Delta h} = \sqrt{(E_{dh})^2 + (1.5)^2 + (0.6)^2} \tag{4.5}$$

During the conversion of volume to mass, the uncertainty in density E_{den} is assumed as ±60 kg/m^3 as suggested by Huss 2013 which is around 7% and the uncertainty in area E_{area} is assumed to be 5 %. This 5% is assumed based on multiple digitization of glacier outlines (Farinotti et al., 2009a). Then the total uncertainty in mass balance calculation is given by:

$$E_{mb} = \sqrt{(E_{\Delta h})^2 + (E_{den})^2 + (E_{area})^2} \tag{4.6}$$

This can also be expressed in relative terms as:

$$\frac{E_{mb}}{mb} = \sqrt{(\frac{E_{\Delta h}}{dh})^2 + (\frac{E_{den}}{density})^2 + (\frac{E_{area}}{area})^2}$$

$$= \sqrt{(\frac{E_{\Delta h}}{dh})^2 + (0.07)^2 + (0.1)^2} \tag{4.6}$$

The value of $\frac{E_{\Delta h}}{dh}$ varies from glacier to glacier.

on the glacier surface. Gardelle et al., 2013 calculated the value of d as lag distance of pixels for which the Moran's I reach $\frac{1}{e}$ i.e., 0.368. The same approach is used in this study and the value of d is calculated to be XXX m for the study between 1999 and 2011 and XXX m for study between 1999 and2013.

The uncertainty in elevation change is also caused due to the penetration correction and seasonality correction. The error due to penetration correction is assumed to be systematic and equal to ±1.5m (Gardelle et al., 2013). The uncertainty due to seasonality correction was assumed 100% that is 0.15 per winter month. In our case it is 4-months summing up to a total error of ±0.6m. All these errors in elevation change are summed up for each glacier using error propagation. This summed up error is denoted as $E_{\Delta h}$

$$E_{\Delta h} = \sqrt{(E_{dh})^2 + (1.5)^2 + (0.6)^2} \tag{4.5}$$

During the conversion of volume to mass, the uncertainty in density E_{den} is assumed as ±60 kg/m^3 as suggested by Huss 2013 which is around 7% and the uncertainty in area E_{area} is assumed to be 5 %. This 5% is assumed based on multiple digitization of glacier outlines (Farinotti et al., 2009a). Then the total uncertainty in mass balance calculation is given by:

$$E_{mb} = \sqrt{(E_{\Delta h})^2 + (E_{den})^2 + (E_{area})^2} \tag{4.6}$$

This can also be expressed in relative terms as:

$$\frac{E_{mb}}{mb} = \sqrt{(\frac{E_{\Delta h}}{dh})^2 + (\frac{E_{den}}{density})^2 + (\frac{E_{area}}{area})^2}$$

$$= \sqrt{(\frac{E_{\Delta h}}{dh})^2 + (0.07)^2 + (0.1)^2} \tag{4.6}$$

The value of $\frac{E_{\Delta h}}{dh}$ varies from glacier to glacier.

5 Results

In this section the results obtained in the present study following the flow chart shown in Figure 4-1 are explained

5.1 Results for Co-registration of DEMs

Following are the results obtained for co-registration of TanDEM-X, SRTM and ASTER DEMs. In the present study TanDEM-X and ASTER DEMs are shifted taking SRTM as the master DEM.

As explained in section 4.3, since the DEMs used in the study are obtained from different sources there is a chance that horizontal and vertical offsets exist. So, co-registration correction is required for these DEMs to minimize the errors due to mis-registration. To apply this correction, we need the stable terrain. Stable terrain in our case has been derived by first delineating the basin boundaries and stream network and then removing the glaciated area and the unstable area near the streams with a buffer of 100m.

The universal co-registration algorithm has been applied and the shift parameters a, b and c are calculated for TanDEM-x DEM and are found to be -16.701, 75.4099 and 0.10225 respectively (Figure 5-1a) and the corresponding shifts in X, Y and Z directions are -16.1624 m, -4.207 m and 0.0544 m respectively. Similarly, the shift parameters a, b and c are calculated for SRTM DEM and are found to be -32.5712, -56.6894 and -2.25664 respectively (Figure 5-1b) and the corresponding shifts in X, Y and Z directions are -27.2199 m, -17.8874 m and -0.9992 m respectively. This shifts in X and Y are used to shift the DEMs accordingly.

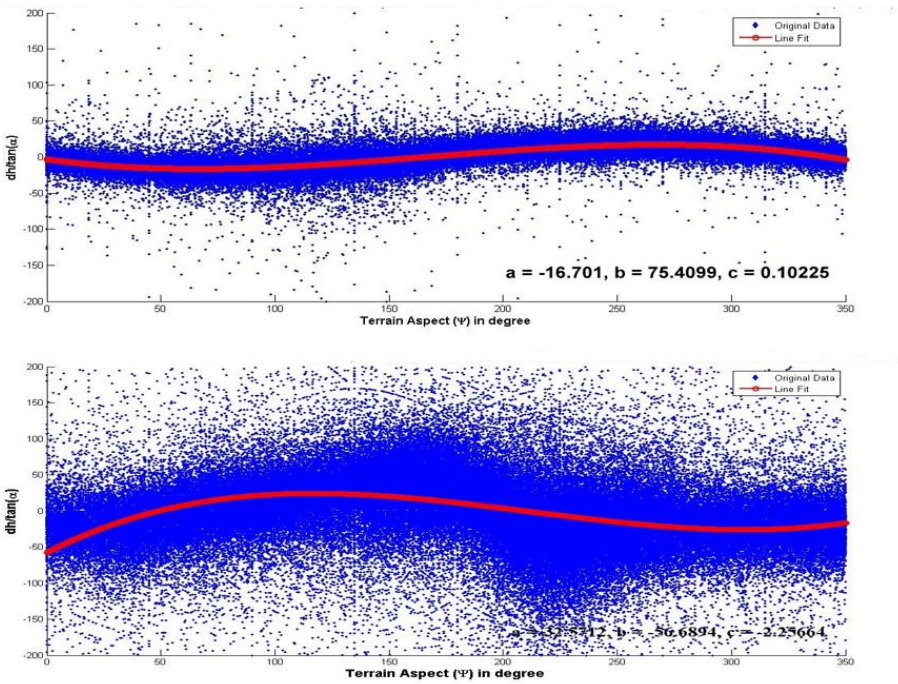

Figure 5-1 The scatter of elevation differences between 2 DEMs showing the relationship between the vertical deviations normalized by the slope tangent (y-axis) and terrain aspect (x-axis). (a) TanDEM-X – SRTM. (b) ASTER - SRTM

5.2 Results for resolution bias correction

Since the DEMs used in this study are of different resolutions (SRTM=30m, TDX=10m) there exists a bias due to different resolutions. This is corrected using the relation between height differences and maximum curvature estimated on stable areas. The stable area is selected in the same way as in section 5.1. The bias between elevation difference and maximum curvature corrected using a first-degree polynomial. The relationship between the altitude and elevation difference before and after correction are shown in Figure 5-2 which shows a clear reduction in bias at higher elevations

Even after the correction of the bias caused by using DEMs of different resolutions there is still bias in elevation changes at higher elevations but this is far less than before correction. This bias cannot be corrected directly as this is not same on and off the glaciers (Gardelle et al. 2012).

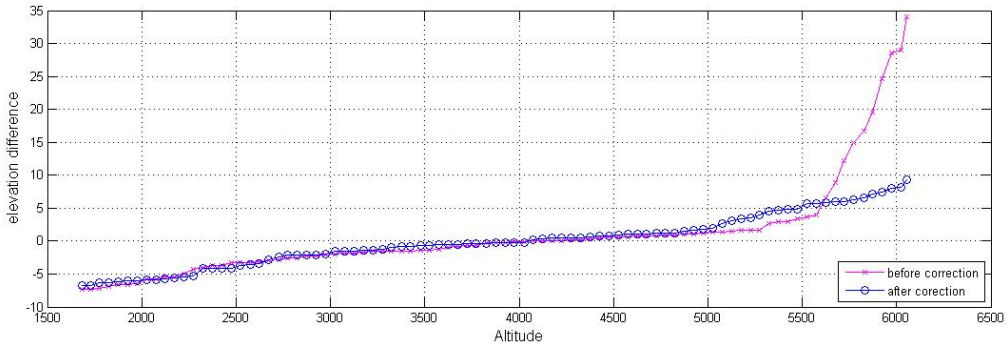

Figure 5-2 Relation between altitude and elevation difference before and after resolution bias correction

After correcting the bias due to resolution change in SRTM DEM we obtain a corrected elevation change map of all the glaciers in Chandra basin between February 2000 and October 2013 (Figure 5-3). From the obtained elevation difference map, the bias corrected SRTM DEM is calculated by adding TanDEM-X DEM to the elevation difference map. The resulting SRTM DEM is of a 10 m resolution.

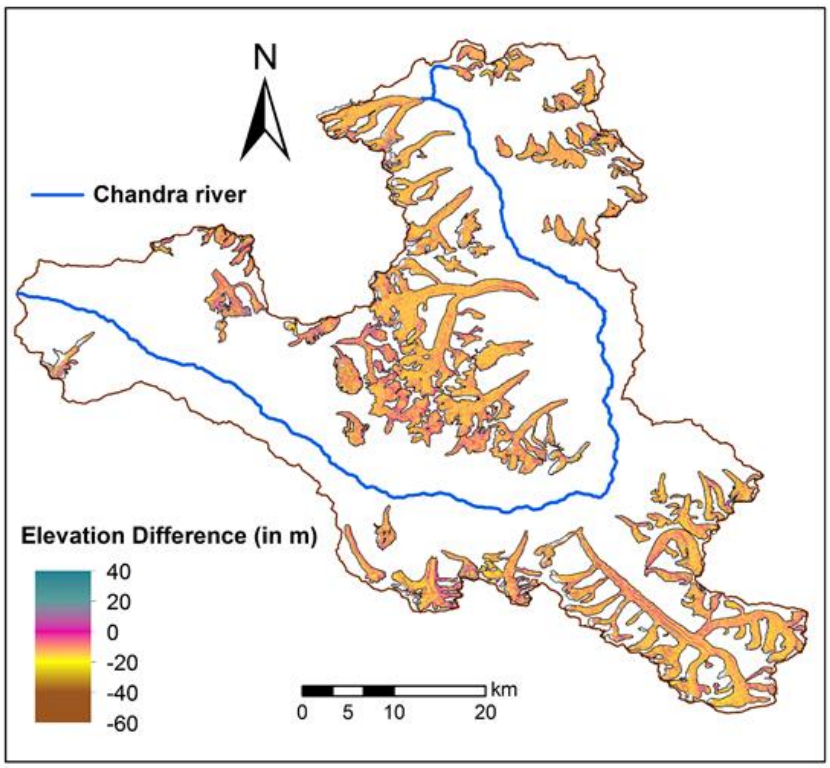

Figure 5-3 elevation change map of all the glaciers in Chandra basin between February 2000 and October 2013

To check the robustness of the corrected SRTM DEM, the corrected SRTM DEM and ASTER DEM are used to calculate the elevation change between 2000 and 2011. The resulting elevation difference map is shown in Figure 5-4

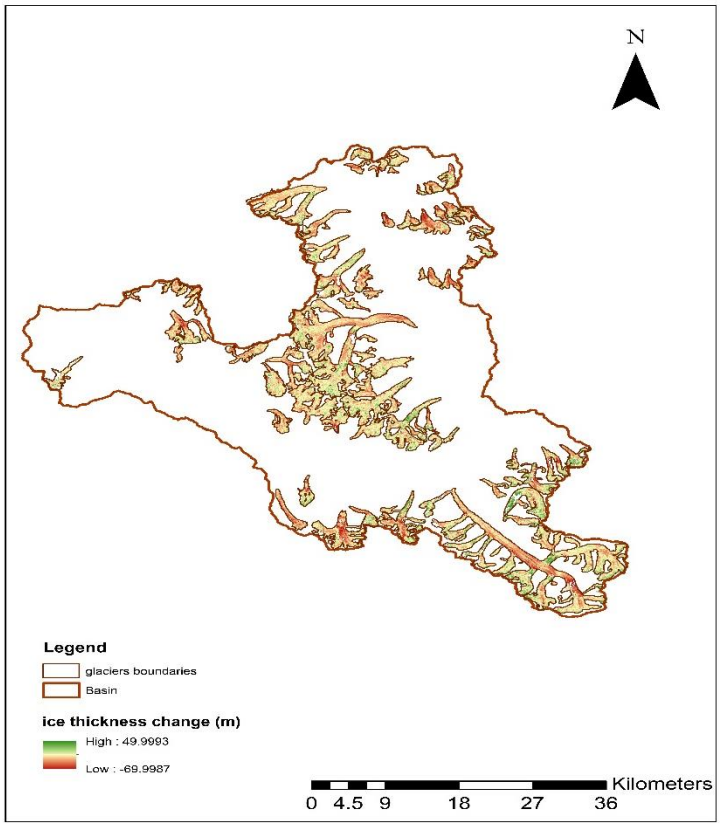

Figure 5-4 elevation change map of all the glaciers in Chandra basin between February 2000 and October 2011

The colour coded ice thickness change maps (Figures 5-3, 5-4) clearly shows that all the glaciers are thinning in most of the parts and a little thickening is observed at accumulation zones of each glacier. The thinning and thickening are found to be varying locally up to 70 m and 50 m respectively.

5.3 Volume change and mass balance 1999 - 2011

The ice thickness change shown in Figure are converted into cumulative mass balance using the ice density value of 850±60 kg/m^3 and to compare the mass balance of different glaciers the specific SMB rate is calculated using the area of individual glacier and the time period. Specific SMB rate of the selected 65 glaciers of Chandra basin obtained using the bias-corrected DEMs between 1999 and 2011 are shown in Figure 5-5 along with the area of each glacier. It is found that the specific SMB rate of different glaciers in Chandra basin range from -1.71 ± 0.31 m w.e.y^{-1} to -0.02 ± .37 m w.e.y^{-1}. The most negative specific SMB rate of -1.71

± 0.31 m w.e.y^{-1} is obtained for the G077610E32652N RGI glacier. The mean specific SMB rate of all the glaciers in Chandra basin is about -0.65 ± 0.22 m w.e.y^{-1}. In comparison with the whole basin it is observed that the mass loss rate is lower in Chhota Shigri Glacier (a bench mark glacier) and Bara Shigri (largest glacier), which are estimated to be -0.53 ± 0.15 m w.e.y^{-1} and -0.54± 0.11 m w.e.y^{-1} respectively. On the other hand, the surface mass loss rate in large glaciers Samudra Tapu is found to more than the rest of the region, which is estimated to be -0.84± 0.13 m w.e.y^{-1}. The basin lost an ice mass of 3.88 ± 0.7 Gt from 1999-2011

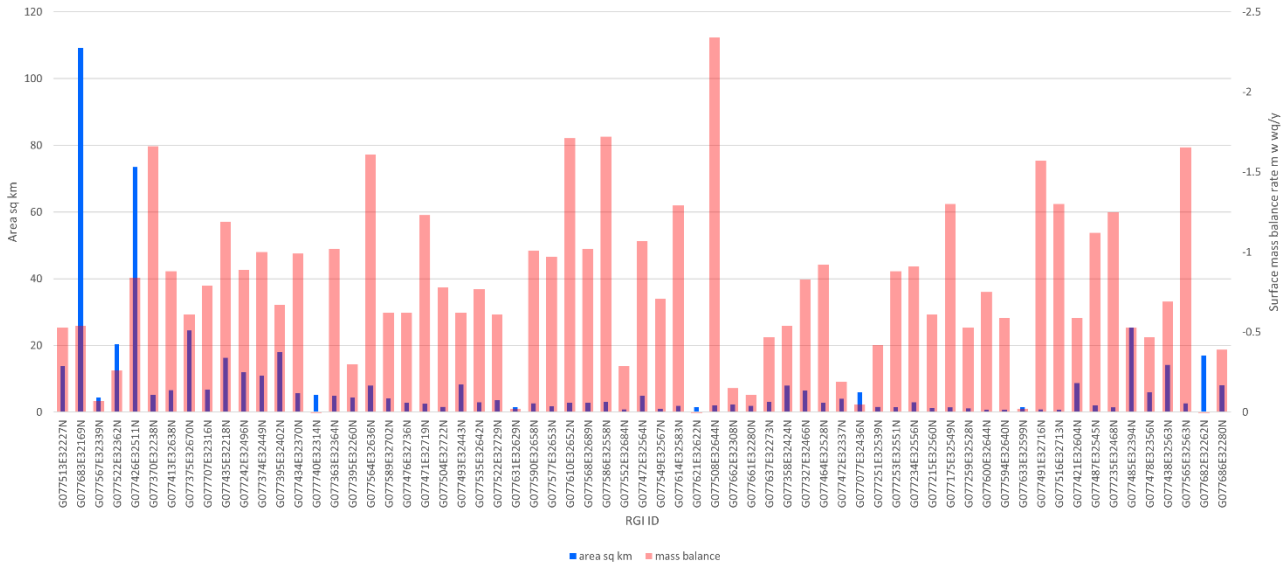

Figure 5-5 Specific mass balance rate of selected 65 glaciers (1999-2011)

5.4 Volume change and mass balance 1999 - 2013

Specific SMB rate of the selected 65 glaciers of Chandra basin obtained using the bias-corrected DEMs between 1999 and 2013 are shown in Figure 5-6 along with the area of each glacier. It is found that the specific SMB rate of different glaciers in Chandra basin range from -0.83 ± 0.15 m w.e.y^{-1} to -0.14 ± .22 m w.e.y^{-1}. The most negative specific SMB rate of -0.83 ± 0.15 m w.e.y^{-1} is obtained for the G077662E32308N RGI glacier. The mean specific SMB rate of all the glaciers in Chandra basin is about -0.61 ± 0.16 m w.e.y^{-1}. In comparison with the whole basin it is observed that the mass loss rate is lower in Chhota Shigri Glacier (a bench mark glacier), which is estimated to be -0.55 ± 0.12 m w.e.y^{-1}. On the other hand, the surface mass loss rate in large glaciers such as Samudra Tapu and Bara Shigri are found to more than

the rest of the region, which are estimated to be -0.69± 0.12 m w.e.y^{-1} and -0.66± 0.11 m w.e.y^{-1} respectively. The basin lost an ice mass of 5.01 ± 0.8 Gt from 1999-2013

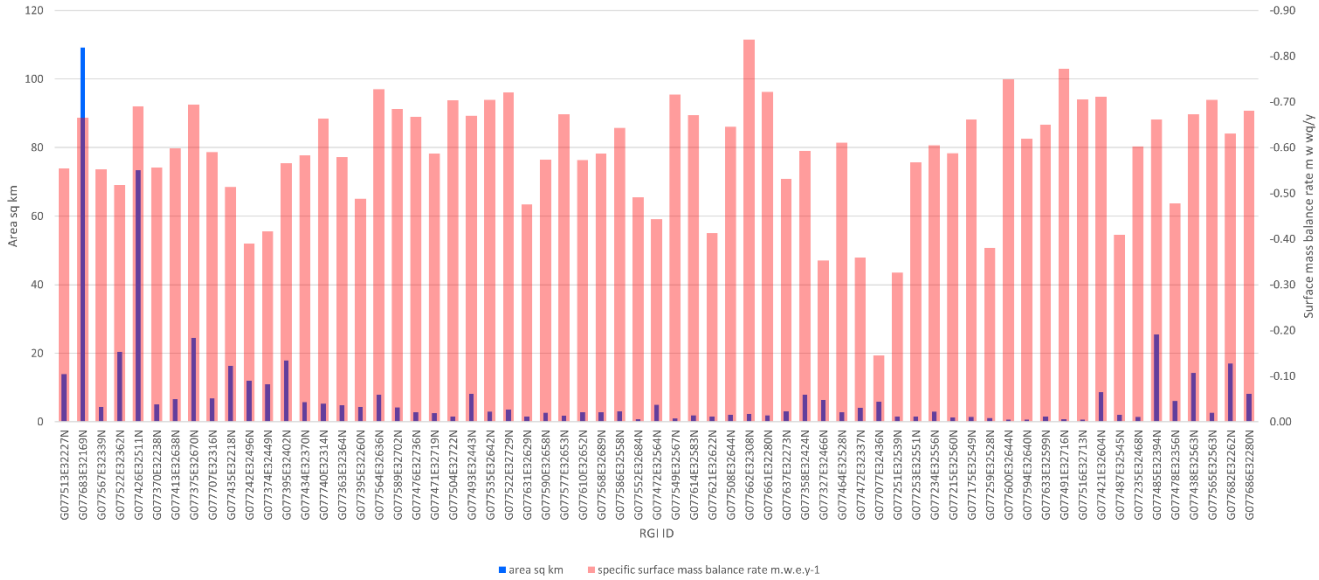

Figure 5-6 Specific mass balance rate of selected 65 glaciers (1999-2013)

5.5 Comparison of results

To quantify the improvements obtained due to DEM bias corrections applied in this study, the specific SMB rate for Chhota Shigri Glacier between 1999-2011 calculated with bias correction (-0.53 ± 0.15 m w.e.y^{-1}) and without bias corrections (0.75± 0.21 m w.e.y^{-1}) are compared with field values (-0.67 m w.e.y^{-1}) obtained from Azam et al. 2016. The comparison indicates that the error has reduced from 211% to 18%. A similar comparison for the study between 1999 and 2013 resulted in an error reduction from 127% to 18%. Therefore, one can say that these corrections are very necessary for better estimation of glacier surface mass balance.

Likewise, a comparison of the mass balance estimates between the present study and other studies given in Table 5-1 indicates that the present study results closely match with the other studies. However, it is observed that the present study estimates for individual glaciers shown in Table 5-1 are slightly overestimated when compared with other studies. This is possibly due to the difference in the methodology used to correct the penetration of C-band. All other mentioned studies have used SRTM-X band data to correct the C-band penetration whereas in

this study the method given by Kaab (2012) has been used. It is mainly because SRTM-X band data for the study area is not available but still other studies have used X-band data of nearby basin as a proxy for correction, which may not be appropriate. Specifically, when the results of Chhota Shigri Glacier (Table 5-1) are compared with the field based glaciological estimates given by Azam et al. 2016, it is found that the present study results are closer than other studies. On the other hand, it is seen that the results of the whole Chandra basin are underestimated compared to other studies (Table5-1). This is because the small glaciers of size <0.5 km^2 are not considered in our study. As smaller glaciers tend to lose more ice compared to larger glaciers the results are under estimated.

Table 5-1 Comparison of Specific SMB rates from present study with other studies in Chandra basin

Study site	**Specific SMB rate from the present study (1999-2011)**	**Specific SMB rate from the present study (1999-2013)**	**Specific SMB rate from other studies**
Chhota Shigri Glacier	-0.53 ± 15 m.w.e.y^{-1}	-0.55 ± 0.12 m.w.e.y^{-1}	#-0.46±0.34 m.w.e.y^{-1}
Bara Shigri Glacier	-0.54 ± 0.11 m.w.e.y^{-1}	-0.66 ± 0.11 m.w.e.y^{-1}	#-0.66±0.31 m.w.e.y^{-1}
Samudra Tapu Glacier	0.84 ± 0.13 m.w.e.y^{-1}	-0.69 ± 0.12 m.w.e.y^{-1}	# -0.69±0.31 m.w.e.y^{-1}
Chandra Basin	-0.65 ± 0.22 m.w.e.y^{-1}	-0.61 ± 0.16 m.w.e.y^{-1}	$-0.68± 0.15 m.w.e.$y^{-1}$; #-0.65± 0.04 m.w.e.$y^{-1}$

Vijay and Braun 2016 (2000-2012), $ Gardelle et al. 2013 (1999-2011), *Vincent et al. 2013 (1999-2011)

6 Conclusions

This thesis explored different approaches to estimate the glacier elevation changes and calculate the glacier surface mass balance. The study is performed on 65 glacier in Chandra basin. The main relevant findings can be summarized as follows:

- Chandra basin lost an ice mass of 3.88 ± 0.7 Gt from 1999-2011.
- Chandra basin lost an ice mass of 5.01 ± 0.8 Gt from 1999-2013.
- Almost all the glaciers in the basin are retreating.
- The surface mass balance rate of Chotta Shigri (Benchmark glacier) is estimated as - 0.53 ± 0.15 m w.e.y^{-1} between 2000 and 2011.
- The surface mass balance rate of Chotta Shigri is estimated as - 0.55 ± 0.12 m w.e.y^{-1} between 2000 and 2013.

This thesis presents a method to estimate the glacier surface mass balance from different DEMs. The results show that the methodology and framework employed in this thesis are effective as the results closely match with field measurements.

Among the possible extensions of the work, some perspectives are:

- Investigation of sources and reasons of elevation change. This thesis quantitatively analysed glacier surface elevation change. Investigation of reason for such changes is meaningful. Potential influencing factors include climate conditions, glacier surges or quiescent dynamics. Indeed, such investigations would be beyond the scope of this thesis, however, are important for glaciological studies.
- Testing with more accurate datasets. Development of new satellite missions such as Resourcesat 3S, ICE Sat-2, CryoSat-2, Sentinel-1 can increase the DEM accuracy, providing better future prospects to understand health of the glaciers.
- Using new technology. Recent technologies such as Unmanned Aerial Vehicle (UAV) are need to be explored which can frequently provide a high-resolution and vertically accurate surface topographic information of the glaciers through which surface mass balance estimates can be obtained on seasonal as well as yearly basis.

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